6.8 Quiz: Polynomials Challenge — Markscheme (b)

1. [Maximum mark: 7]

   Consider the function $f(x) = -2(x+3)(x-5)$, for $x \in \mathbb{R}$.

   (a)(i) Write down the x-intercepts. [2]

   $x = -3$ and $x = 5$ **A1A1**

   (a)(ii) Find the coordinates of the vertex. [3]

   Axis of symmetry: $x = \dfrac{-3+5}{2} = 1$ **M1**

   M1 may also be awarded for any attempt at completing the square.

   $f(1) = -2(1+3)(1-5) = -2(4)(-4) = 32$

   Vertex: $(1, 32)$ **A1A1**

   (b) $f(x) = -2(x-h)^2 + k$. Write down h and k. [2]

   $h = 1$ **A1**

   $k = 32$ **A1**

   FT: award A1A1 for h and k read from the candidate's vertex in (a)(ii).

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2. [Maximum mark: 7]

   The line $y = mx$ is tangent to the parabola $y = x^2 + 16$.

   (a) Show that at any intersection, $x^2 - mx + 16 = 0$. [1]

   $x^2 + 16 = mx$ **M1**

   $x^2 - mx + 16 = 0$ **AG**

   (b) Find the discriminant in terms of m. [1]

   $\Delta = (-m)^2 - 4(1)(16) = m^2 - 64$ **A1**

   (c) Find the values of m for tangency. [2]

   Tangent: $\Delta = 0$ **M1**

   $m^2 - 64 = 0$

   $m = \pm 8$ **A1**

   (d) Find the coordinates of each point of tangency. [3]

For $m = 8$: $x^2 - 8x + 16 = 0$

$(x-4)^2 = 0$ **M1**

$x = 4$

$y = 8(4) = 32$. Point of tangency: $(4, 32)$ **A1**

For $m = -8$: $x^2 + 8x + 16 = 0$

$(x+4)^2 = 0$

$x = -4$

$y = -8(-4) = 32$. Point of tangency: $(-4, 32)$ **A1**

FT: award A1 for each point correctly computed from the candidate's x-value.

3. [Maximum mark: 7]

Consider the function $f(x) = 3x^2 + 6x - 2$.

(a) Write $f(x)$ in the form $a(x-h)^2 + k$. [3]

$f(x) = 3(x^2 + 2x) - 2$ **M1**

M1 for any attempt at completing the square, including use of $-b/(2a)$ anywhere.

$= 3(x^2 + 2x + 1 - 1) - 2 = 3(x+1)^2 - 3 - 2$

$= 3(x+1)^2 - 5$ **A1A1**

(b) Write down the coordinates of the vertex. [1]

Vertex: $(-1, -5)$ **A1**

(c) Find the discriminant of $f(x) = 0$. [1]

$\Delta = (6)^2 - 4(3)(-2) = 36 + 24 = 60$ **A1**

(d) State the number of x-intercepts and justify. [2]

Two x-intercepts. **A1**

Accept "two", "2", "double", or equivalent.

Justification: $\Delta = 60 > 0$, so two distinct real roots. **R1**

R1 may also be awarded for naming the two roots, or for stating that the candidate's $\Delta > 0$.

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4. [Maximum mark: 8]

Let $p(x) = 2x^3 - 3x^2 - 11x + 6$.

(a) Show that $(x-3)$ is a factor of $p(x)$. [2]

$p(3) = 2(27) - 3(9) - 11(3) + 6$ **M1**

$= 54 - 27 - 33 + 6 = 0$

Since $p(3) = 0$, $(x-3)$ is a factor by the factor theorem. **A1 AG**

(b) Express $p(x) = (x-3)(ax^2 + bx + c)$. [2]

By polynomial division: **M1**

$2x^3 - 3x^2 - 11x + 6 = (x-3)(2x^2 + 3x - 2)$ **A1**

Accept $a = 2$, $b = 3$, $c = -2$.

M1 may also be awarded for synthetic division (Ruffini's rule).

(c) Factorise $p(x)$ completely. [2]

$2x^2 + 3x - 2 = (2x - 1)(x + 2)$ **M1**

M1 may also be awarded for use of the quadratic formula.

$p(x) = (x - 3)(2x - 1)(x + 2)$ **A1**

Accept $2(x - \frac{1}{2})$ *in place of* $(2x - 1)$*; do not accept* $(x - \frac{1}{2})$ *alone (missing leading coefficient).*

(d) Write down all solutions to $p(x) = 0$. [1]

$x = 3, \quad x = \dfrac{1}{2}, \quad x = -2$ **A1**

(e) Write down the y-intercept. [1]

$p(0) = 6$, so the y-intercept is $(0, 6)$. **A1**